

Symmetric-Key Algorithms

Encryption in which the parties **share** a secret key

→ The **same** key!

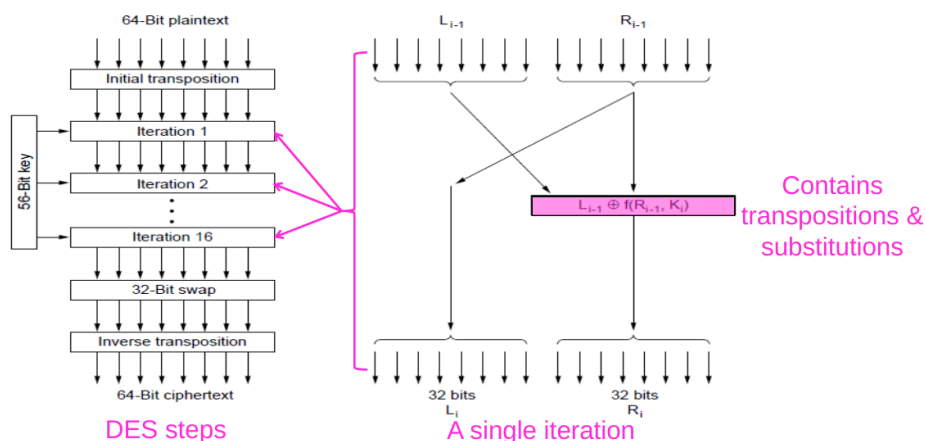
Outline

- DES – Data Encryption Standard
- Cipher modes (cipher block chaining)

DES: Data Encryption Standard

Fundamentals:

- **Symmetric key algorithm: encryption and decryption with the same key**
- **Plaintext encrypted in blocks of 64 bits**
- **19 stages**
- **56-bit key (other versions exists with longer keys); the 56-bit is not any more secure**



SUBSTITUTION CIPHER

Ogni simbolo si sostituisce con un altro.
La chiave è la corrispondenza tra simboli.
Per alfabeto romano (e altri) è facilmente attaccabile.

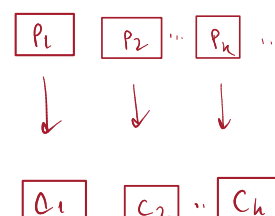
Le regolarità nel testo in chiaro
si riflettono nel testo cifrato!

- First and last stage: transposition (one inverse of the other)
- 18th stage: swap of the two 32-bit blocks (see figure)

Intermediate stages

- Left 32 bits L_{i-1} unchanged
- E is an expansion/transposition of R_{i-1}
- E is XORed with K_i
- K_i is different for each i (transposition)

Il DES eseguito blocco a blocco non è altro che una sostituzione di simboli in altri simboli (cifatura per sostituzione), perché ogni blocco da 64bit viene crittato separatamente. (ovviamente quello qui sopra non è il DES che viene usato in pratica; v. cipher block chaining qui sotto).

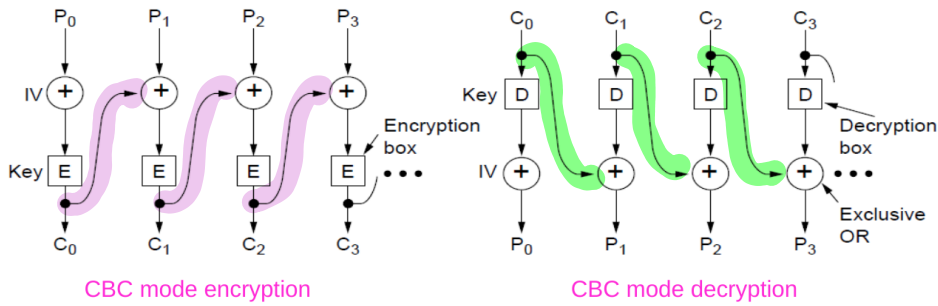


$S_1 S_1 S_2 \dots S_1$
 $S_3 S_3 S_3 \dots S_3$

Cipher Modes

CBC (Cipher Block Chaining) is a widely used mode

- Chains blocks together with XOR to prevent shifts
- Has a random IV (Initial Value) for different output



Public-Key Algorithms

Encryption in which each party publishes a **public** part of their key and keep secret a **private** part of it

- **RSA** (by Rivest, Shamir, Adleman)

Public-Key Algorithms

Downsides of keys for symmetric-key designs:

- Key must be secret, yet be distributed to both parties
- For N users there are N^2 pairwise keys to manage

Public key schemes split the key into public and private parts that are mathematically related:

- Private part is not distributed; easy to keep secret
- Only one public key per user needs to be managed

Security depends on the chosen mathematical property

- Much slower than symmetric-key, e.g., 1000X
- So use it to set up per-session symmetric keys

Public Key algorithms: requirements

1. $D(E(P)) = P$ (correctness);
2. It is extremely difficult to deduce the private key from the public key;
3. E cannot be broken by a **chosen plaintext attack**

l'attacco è in grado di analizzare codice arbitrario

< testata in chiaro, Testo Cifrato >

RSA (optional slide)

RSA is a widely used public-key encryption method whose security is based on the difficulty of factoring large numbers

Key generation:

- Choose two large primes, p and q
- Compute $n = p \times q$ and $z = (p - 1) \times (q - 1)$.
- Choose d to be relatively prime to z
- Find e such that $e \times d = 1 \text{ mod } z$
- Public key is (e, n), and private key is (d, n)

Encryption (of k bit message, for numbers up to n):

- $\text{Cipher} = \text{Plain}^e \text{ (mod } n)$

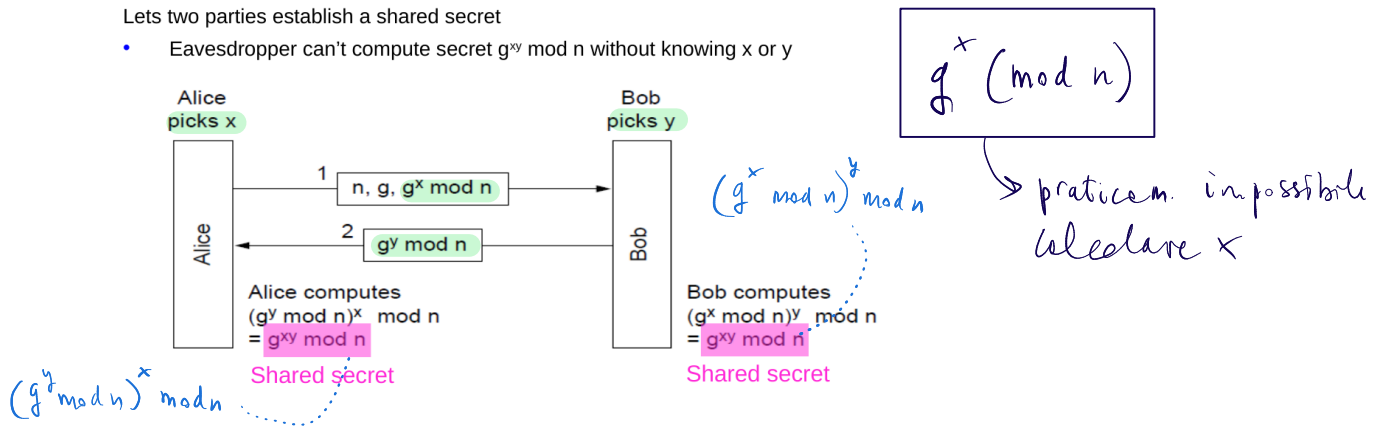
Decryption:

- $\text{Plain} = \text{Cipher}^d \text{ (mod } n)$

Diffie-Hellman Key Exchange

Lets two parties establish a shared secret

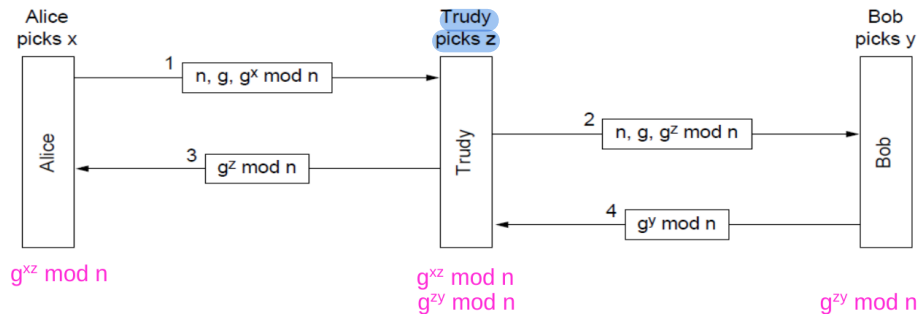
- Eavesdropper can't compute secret $g^{xy} \bmod n$ without knowing x or y



Diffie-Hellman Key Exchange: attack

But it is vulnerable to a **man-in-the-middle** attack

- Need to confirm identities, not just share a secret



Four keys can be created from the MS:

- E_{B-to-A} = session **encryption key** for data sent from B to A
- M_{B-to-A} = session **MAC key** for data sent from B to A
- E_{A-to-B} = session **encryption key** for data sent from A to B
- M_{A-to-B} = session **MAC key** for data sent from A to B

MAC: Message Authentication Code

WHonix (Ton)